



2024

J20

BOOKLET NO.

320276

**Forest Services
Forestry**

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – 1. Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

SEAL

**SECTION – A**

Q1. Write short notes on **any five** from seven. **(8×5=40)**

- (a) Write in detail about global warming and its effect.
- (b) Comment on distribution and economic importance of *Acacia catechu*.
- (c) Define desert. Write a note on cold desert flora.
- (d) Write a note on forest nursery in respect of permanent and temporary nurseries.
- (e) Write in brief about Mycorrhizae.
- (f) Discuss in brief about criteria and indicators of sustainable forest management.
- (g) Write in brief about seed production area and seed orchards.

-
- Q2.** (a) Give the classification of high forest systems and explain the selection system in detail. **15**
- (b) Discuss the phenology, silvicultural characters and regeneration methods of *Anogeissus latifolia*. **15**
- (c) What are mangroves ? Write in short about unique features of mangrove plants. **10**

-
- Q3.** (a) Describe the structural classification of Agroforestry systems. **15**
- (b) Elaborate visual symptoms of deficiency of macro and micro nutrient Elements in detail. **15**
- (c) What do you understand by Environment ? Explain components of Environment. **10**

-
- Q4.** (a) Discuss the methods of Conservation of forest genetic resources. **15**
- (b) Explain normal forest and growing stock. Discuss relationship between current and mean annual increment of forest tree. **15**
- (c) Discuss in brief the stages of preparation of working plan. **10**
-

**SECTION – B**

Q5. Solve **any five** from seven.

(8×5=40)

- (a) Write about India's applications of remote sensing in Forestry ?
 - (b) Write about Forests types of India.
 - (c) Fire is a major bane of forests. Explain with examples.
 - (d) Write about various initiatives for tribal welfare through Forest department.
 - (e) How does forest leaf litter play an important role in carbon cycle ?
 - (f) What are Biodiversity hotspots ? Write about Biodiversity hotspots in India.
 - (g) Responsible wood sourcing is a step towards sustainability – Explain.
-

- Q6.** (a) Write about the methods of diameter, girth, height and volume measurement in tress. Enlist applications of these in forestry. **15**
- (b) Briefly explain different types of composite wood. In your opinion, what is the status of this industry in India ? **15**
- (c) Write in detail about trends and patterns in use of wood in the world. **10**
-

- Q7.** (a) Explain ecological succession. How is succession important for forest management ? **15**
- (b) Explain the concept of territoriality in wild animals with appropriate examples in mammals and birds. **15**
- (c) What are the similarities and differences between Ayurveda and Unani systems of medicine ? **10**
-

- Q8.** (a) Write about forestry and allied activities used for watershed management. **15**
- (b) How forest policy and legislation helped to conserve forests and wildlife in India ? **15**
- (c) What is a map ? Describe its components. What are different maps used in forestry ? **10**
-

SEAL